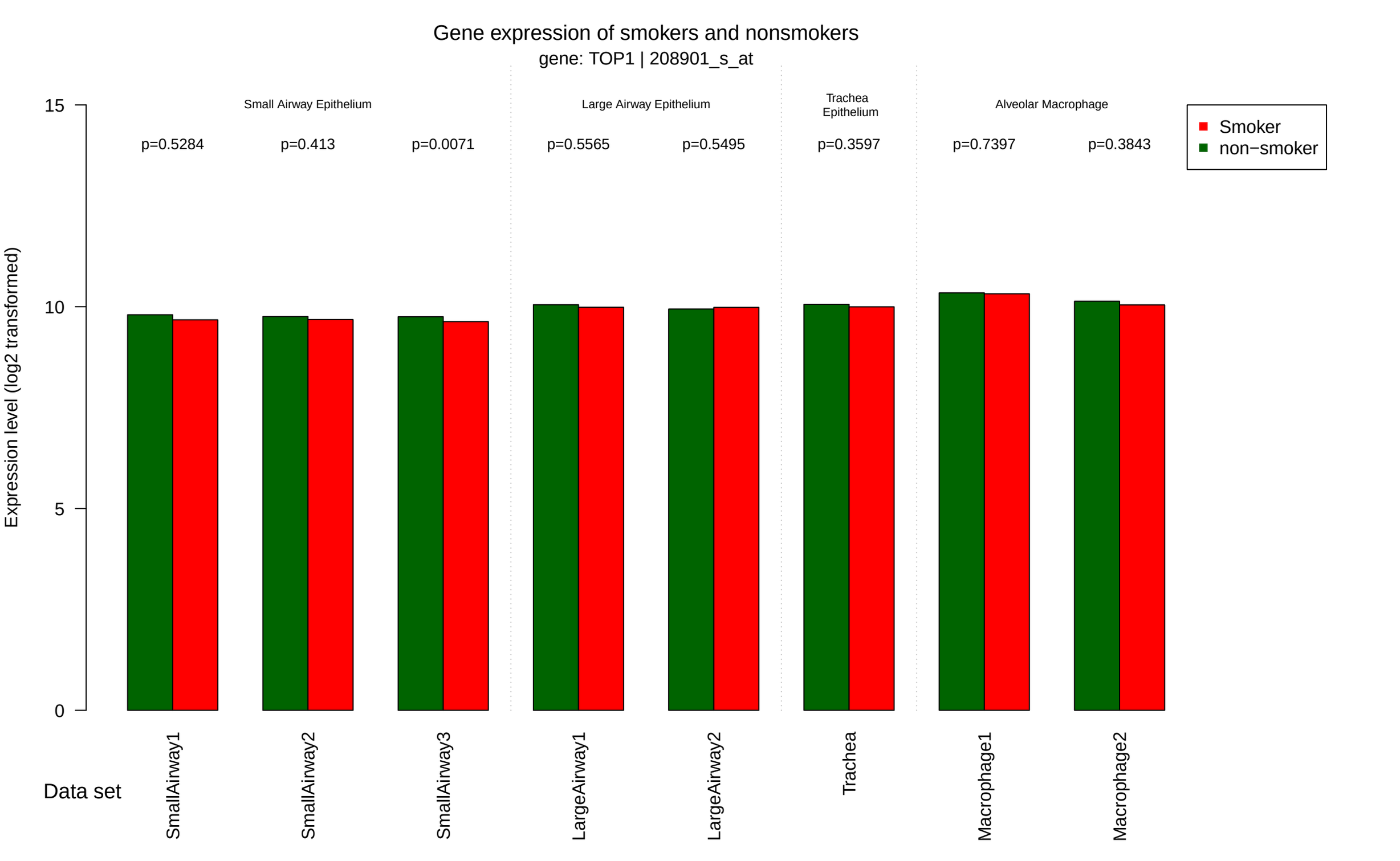
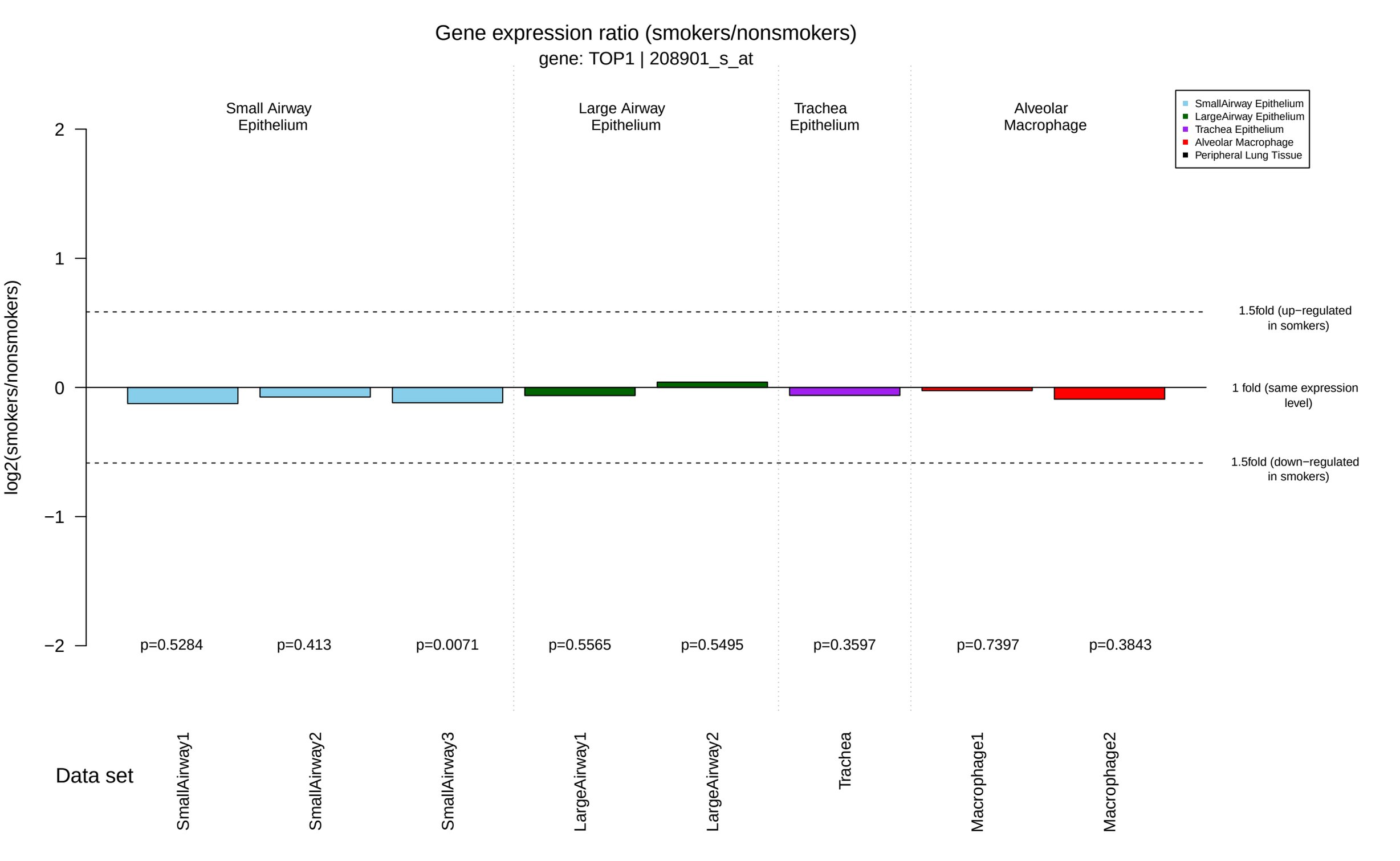
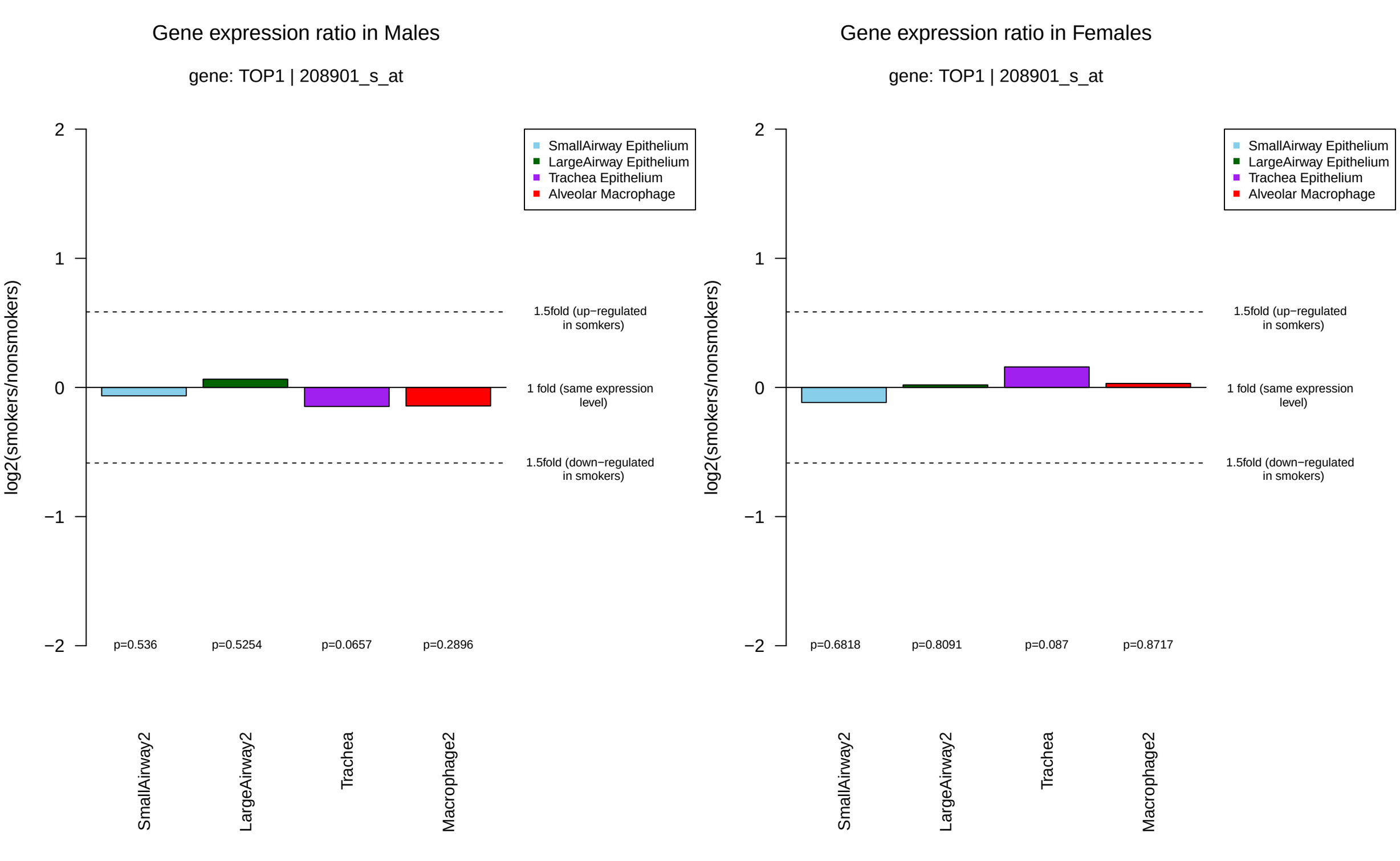
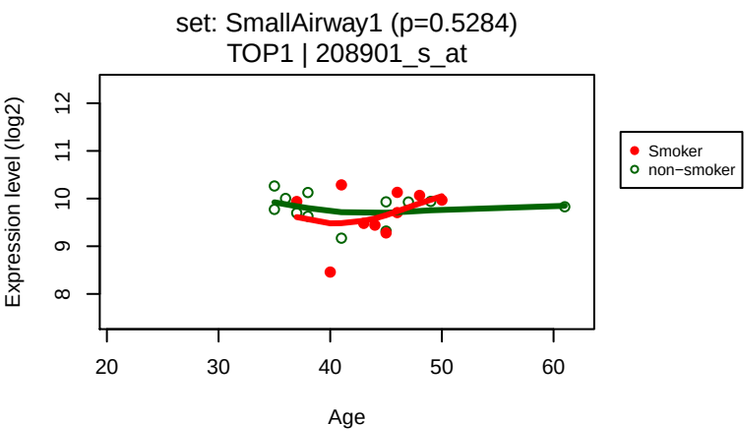
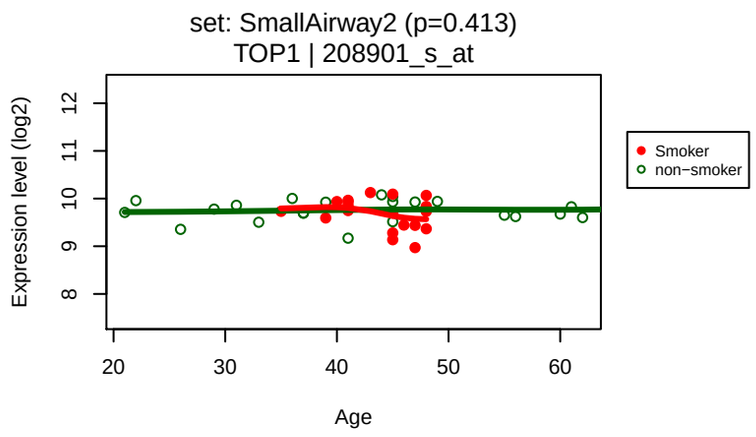
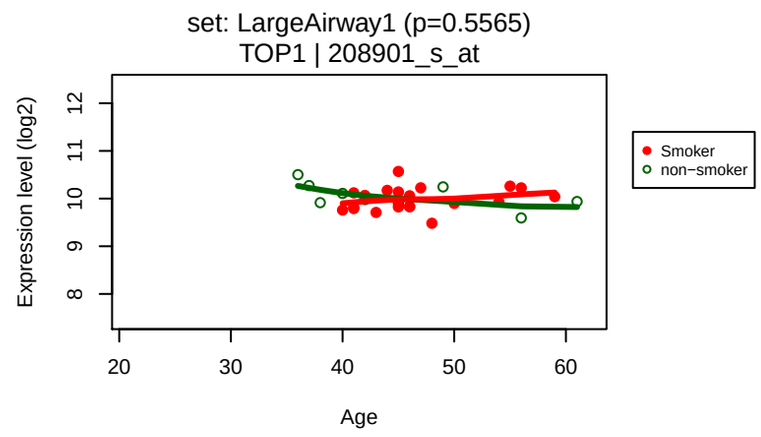

Gene expression of smokers and nonsmokers
gene: TOP1 | 208901_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5284
p=0.413
p=0.0071
p=0.5565
p=0.5495
p=0.3597
p=0.7397
p=0.3843
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

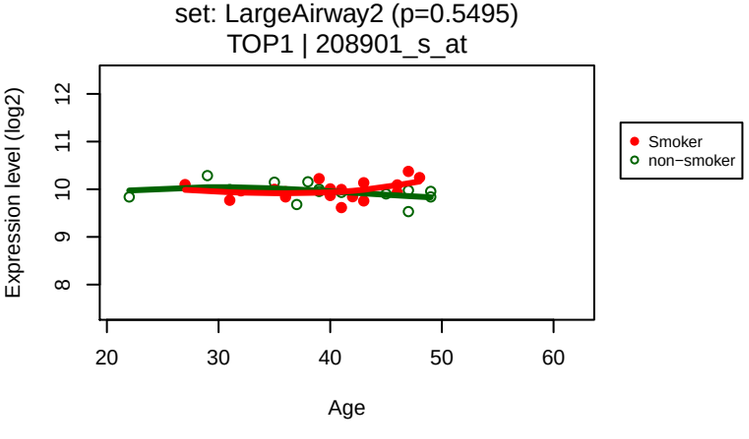

Gene expression ratio (smokers/nonsmokers)
gene: TOP1 | 208901_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5284
p=0.413
p=0.0071
p=0.5565
p=0.5495
p=0.3597
p=0.7397
p=0.3843
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

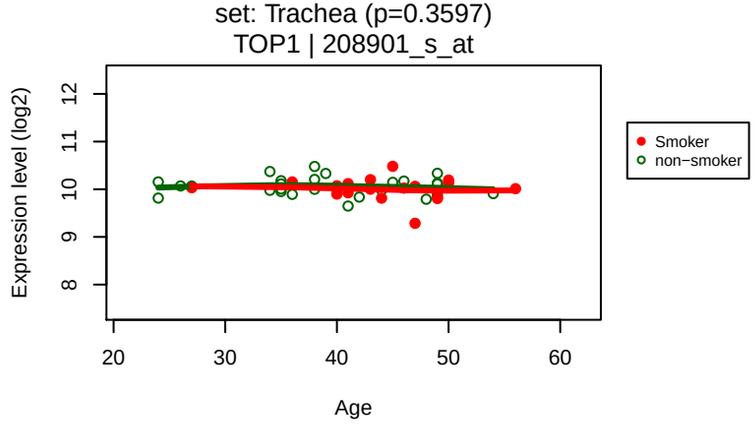

Gene expression ratio in Males
gene: TOP1 | 208901_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.536
p=0.5254
p=0.0657
p=0.2896
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TOP1 | 208901_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6818
p=0.8091
p=0.087
p=0.8717
SmallAirway2
LargeAirway2
Trachea
Macrophage2

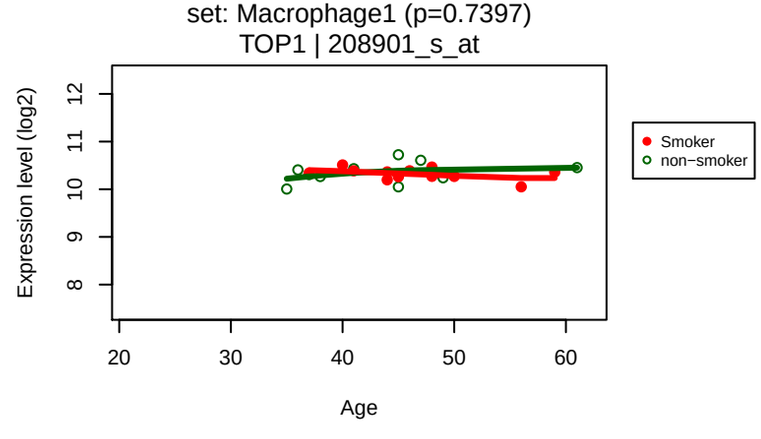

set: SmallAirway1 (p=0.5284)
TOP1 | 208901_s_at
Expression level (log2)
Age
Smoker
non−smoker

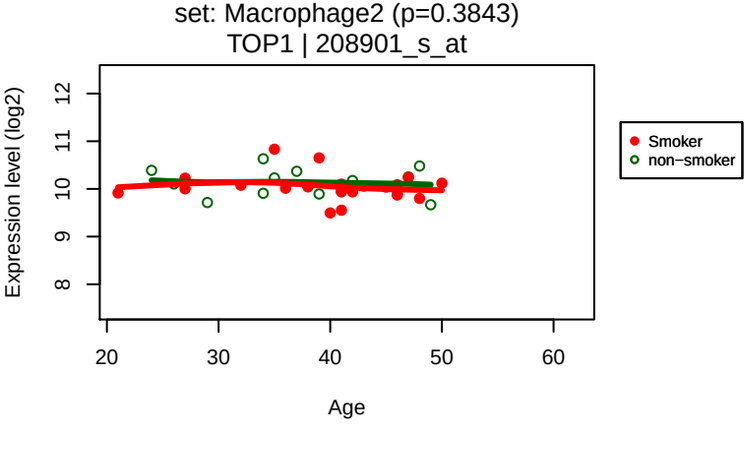

set: SmallAirway2 (p=0.413)
TOP1 | 208901_s_at
Expression level (log2)
Age
Smoker
non−smoker

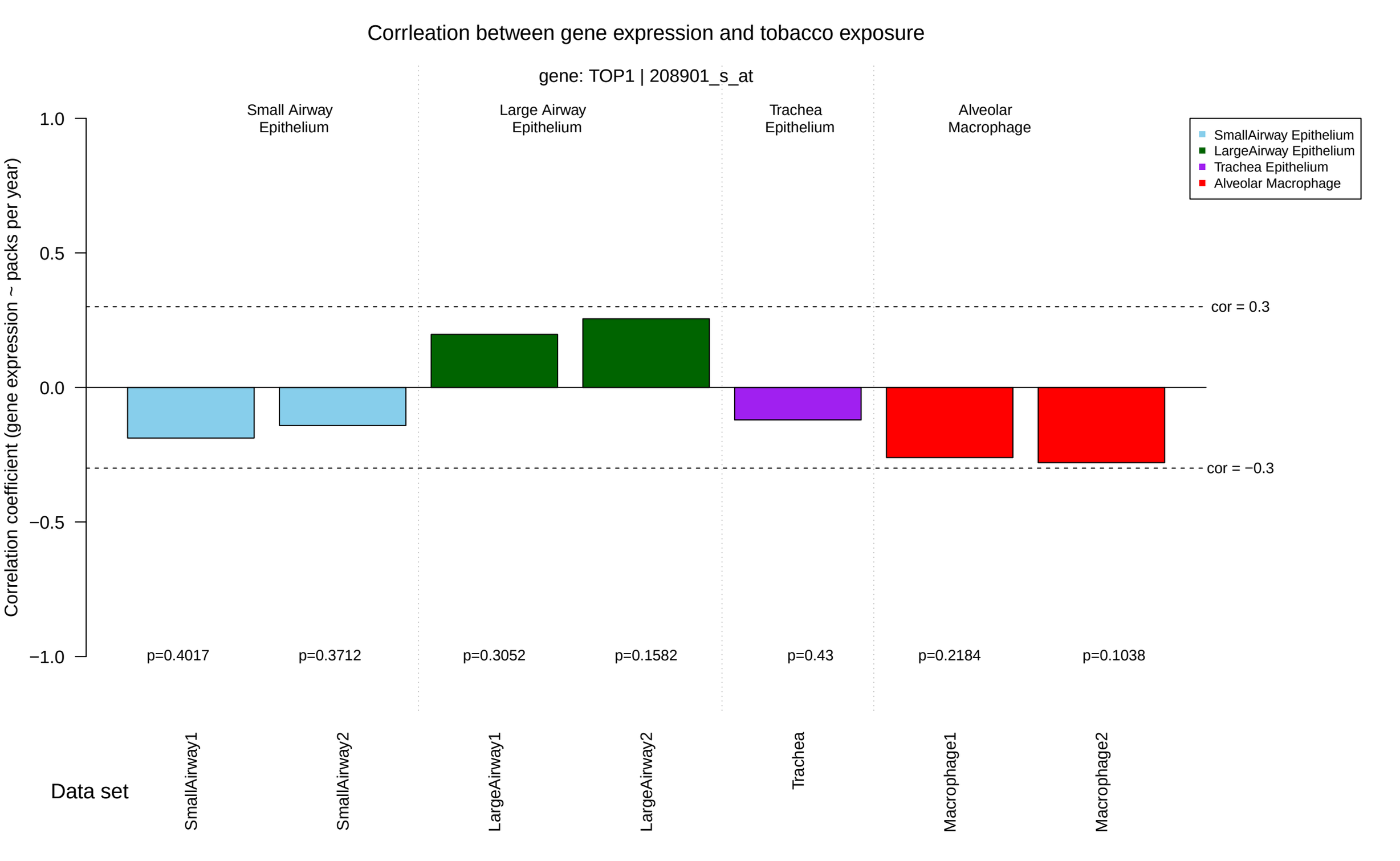

set: LargeAirway1 (p=0.5565)
TOP1 | 208901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5495)
TOP1 | 208901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3597)
TOP1 | 208901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7397)
TOP1 | 208901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3843)
TOP1 | 208901_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TOP1 | 208901_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4017
p=0.3712
p=0.3052
p=0.1582
p=0.43
p=0.2184
p=0.1038
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2